Cambridge University Engineering Department
Engineering Tripos Part IIA
PROJECTS: Interim and Final Report Coversheet

TO BE COMPLETED BY THE STUDENT(S)

Project:	SA1 - Wing Analysis
Title of report:	Group Report / ~~Individual Report~~ (delete as appropriate)

Name(s): (capitals)	crsID(s):	College(s):
Joe Parker	jp2056	Clare
Gereon Leckebusch	gtl23	Girton

Declaration for: ~~Interim Report 1~~ / Interim Report 2 / ~~Final Report~~ (delete as appropriate)

I/we confirm that, except where indicated, the work contained in this report is my/our own original work.

Instructions to markers of Part IIA project reports:

Grading scheme

Grade	A*	A	B	C	D	E
Standard	Excellent	Very Good	Good	Acceptable	Minimum acceptable for Honours	Below Honours

Grade the reports by ticking the appropriate guideline assessment box below, and provide feedback against as many of the criteria as are applicable (or add your own). Feedback is particularly important for work graded C-E. Students should be aware that different projects and reports will require different characteristics.

Penalties for lateness: Interim Reports: 3 marks per weekday; Final Reports: 0 marks awarded – late reports not accepted.

Guideline assessment (tick one box)

A*/A	**A/B**	**B/C**	**C/D**	**D/E**

Marker:		Date:	

Delete (1) or (2) as appropriate (for marking in hard copy – different arrangements apply for feedback on Moodle):

(1) Feedback from the marker is provided on the report itself.

(2) Feedback from the marker is provided on second page of cover sheet.

	Typical Criteria	Feedback comments
Project Skills, Initiative, Originality	Appreciation of problem, and development of ideas	
	Competence in planning and record-keeping	
	Practical skill, theoretical work, programming	
	Evidence of originality, innovation, wider reading (with full referencing), or additional research	
	Initiative, and level of supervision required	
Report	Overall planning and layout, within set page limit	
	Clarity of introductory overview and conclusions	
	Logical account of work, clarity in discussion of main issues	
	Technical understanding, competence and accuracy	
	Quality of language, readability, full referencing of papers and other sources	
	Clarity of figures, graphs and tables, with captions and full referencing in text	

SA1 - Week 2 Report

Joe Parker, Gereon Leckebusch

May 2026

1 Zero-pressure-gradient, laminar boundary layer

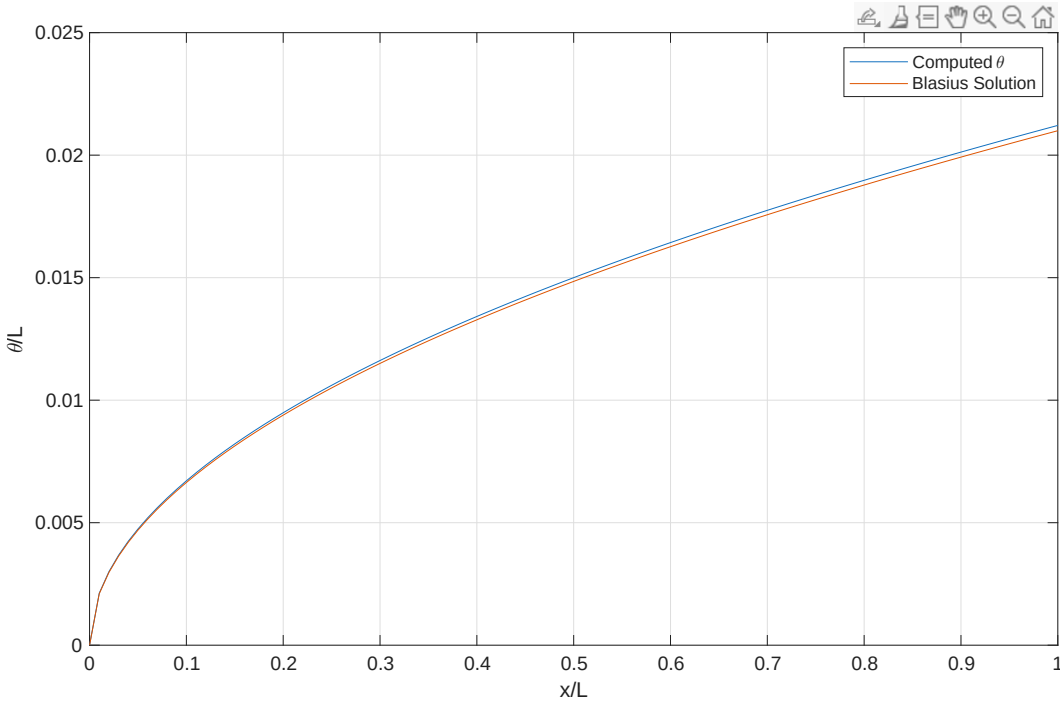


Figure 1: Momentum thickness against distance for a flat plate boundary layer in zero pressure gradient. Both the Blasius solution and numerical solution are plotted. (Exercise 1)

Listing 1: The function to evaluate the contribution to the boundary layer integral. (Exercise 1)

```
function f = ueintbit(xa, ua, xb, ub)
    ubar = (ua+ub)/2;
    du = ub-ua;
    dx = xb-xa;
    f = (ubar^5 + (5/6)*ubar^3 * du^2 + (1/16) * ubar * du^4)*dx;
end
```

Listing 2: The script for plotting the boundary layer thickness of a flat plate with zero pressure gradient. (Exercise 1)

```
np = 101;
x = linspace(0,1,np);
u_ = 1*ones(1,np); %dimensionless velocity

int = 0; % value of the integral
theta_2 = zeros(1,np);  %theta squared at each position
re_l = 1000;
for i=2:np
    int = int + ueintbit(x(i-1), u_(i-1), x(i), u_(i));
    theta_2(i) = (0.45/re_l) * u_(i)^(-6) * int;
```

```
end
analy = 0.664/re_l^(0.5) * x.^0.5;
theta = sqrt(theta_2);


figure;
plot(x, theta);
hold on;
plot(x, analy);
hold off;
xlabel('x/L');
ylabel('\theta/L');
legend('Computed \theta', 'Blasius Solution');
title('Comparison of Computed and Analytical Theta');
grid on;
print -deps2c Week_2/Exercise1.eps;
```

2 Constant-velocity-gradient, laminar boundary layer

	x/L at transition		
Re_L \ du/dx	-0.1	0	0.1
5×10^6	0.48	0.74	None
10×10^6	0.29	0.37	0.55
20×10^6	0.17	0.19	0.22

Table 1: Transition location for varying Reynolds numbers and velocity gradients.

	Re_θ at transition		
Re_L \ du/dx	-0.1	0	0.1
5×10^6	1080	1290	None
10×10^6	1168	1290	1515
20×10^6	1253	1308	1385

Table 2: Re_θ at transition locations for varying Reynolds numbers and velocity gradients.

Listing 3: The script for finding the transition location and Reynold's numbers for a flat plate boundary layer with a constant velocity gradient (Exercise 2).

```
clear;
close all;

np = 101;
laminar = true;
x = linspace(0,1,np); %length varies from 0->1
dudx = 0; %dimensionless gradient

integ = 0; % value of the integral
theta_2 = zeros(1,np);  %theta squared at each position
re_ls=[5e6,10e6,20e6]; %searching for transition
dudxs = [-0.1,0,0.1];

for n=1:length(re_ls)

    for k=1:length(dudxs)
        re_l = re_ls(n);
        dudx = dudxs(k);
        laminar = true;
```

```
        integ = 0;
        int = 0;
        ils = 0;
        i=1;
        u_ = linspace(1, 1+dudx, np);
        while laminar && i<np
            i=i+1;
            integ = integ + ueintbit(x(i-1), u_(i-1), x(i), u_(i));
            theta_2(i) = (0.45/re_l) * u_(i)^(-6) * integ;
            m = -re_l * theta_2(i)  * dudx;
            H = thwaites_lookup(m);
            He = laminar_He(H);
            re_thet = re_l * u_(i) * sqrt(theta_2(i));
            if log(re_thet) >= 18.4*He - 21.74
                laminar = false;    %transition
                int = i;
        end
        if int ~= 0
            found =true;
            disp(['re_l = ' num2str(re_l) ', dudx = ' num2str(dudx)])
            disp(['Natural transition at x = ' num2str(x(int)) ', Rethet/1000 = ' num2str(re_l * u_(int) * sqrt
    (theta_2(int)) / 1000)])
        else
            disp(['re_l = ' num2str(re_l) ', dudx = ' num2str(dudx)])
            disp('No transition')
        end
        disp(' ')
    end
end
```

3 Laminar boundary layer with transition or separation

For transition to occur before separation at a velocity gradient of $du/dx = -0.25$, the Reynolds number needs to be $Re_L = 9.0 \times 10^5$.

Reynold's Number	Separation Location
1×10^5	0.50
1×10^4	0.50
1×10^3	0.50

Table 3: The separation locations, x/L, for laminar boundary layers with a constant velocity gradient $du/dx = -0.25$. (Exercise 3)

Listing 4: The script for finding the separation or transition location of a laminar boundary layer. It is currently set up to find the Reynolds number at which transition supplants separation. (Exercise 3)

```
clear;
close all;

np = 101;
laminar = true;
x = linspace(0,1,np); %length varies from 0->1
dudx = 0; %dimensionless gradient

integ = 0; % value of the integral
theta_2 = zeros(1,np);  %theta squared at each position
re_ls=linspace(1e5,1e6,10000); %searching for transition Reynold's number
dudxs = [-0.25];
found = false;
for n=1:length(re_ls)
    if found
        break
```

```
    end
    for k=1
        re_l = re_ls(n);
        dudx = dudxs(k);
        laminar = true;
        integ = 0;
        int = 0;
        ils = 0;
        i=1;
        u_ = linspace(1, 1+dudx, np);
        while laminar && i<np
            i=i+1;
            integ = integ + ueintbit(x(i-1), u_(i-1), x(i), u_(i));
            theta_2(i) = (0.45/re_l) * u_(i)^(-6) * integ;
            m = -re_l * theta_2(i)  * dudx;
            H = thwaites_lookup(m);
            He = laminar_He(H);
            re_thet = re_l * u_(i) * sqrt(theta_2(i));
            if log(re_thet) >= 18.4*He - 21.74
                laminar = false;    %transition
                int = i;

            elseif m>= 0.09
                laminar = false;
                ils = i;
            end
        end
        if int ~= 0
            found =true;
            disp(['re_l = ' num2str(re_l) ', dudx = ' num2str(dudx)])
            %disp([num2str(re_ls(n-1)) ', ' num2str(re_l) ', ' num2str(re_l) ', ' num2str(re_ls(n+1))])
    %surrounding reynolds numbers to check we have enough sig figs
            disp(['Natural transition at x = ' num2str(x(int)) ', Rethet/1000 = ' num2str(re_l * u_(int) * sqrt
    (theta_2(int)) / 1000)])
        elseif ils ~= 0
            disp(['re_l = ' num2str(re_l) ', dudx = ' num2str(dudx)])
            %disp([num2str(re_ls(n-1)) ', ' num2str(re_l) ', ' num2str(re_l) ', ' num2str(re_ls(n+1))])

            disp(['Separation at x = ' num2str(x(ils)) ', Rethet/1000 = ' num2str(re_l * u_(ils) * sqrt(theta_2
    (ils)) / 1000)])
        else
            disp(['re_l = ' num2str(re_l) ', dudx = ' num2str(dudx)])
            disp('No transition or separation')
        end
        disp(' ')
```

4 Zero-pressure-gradient, turbulent boundary layer

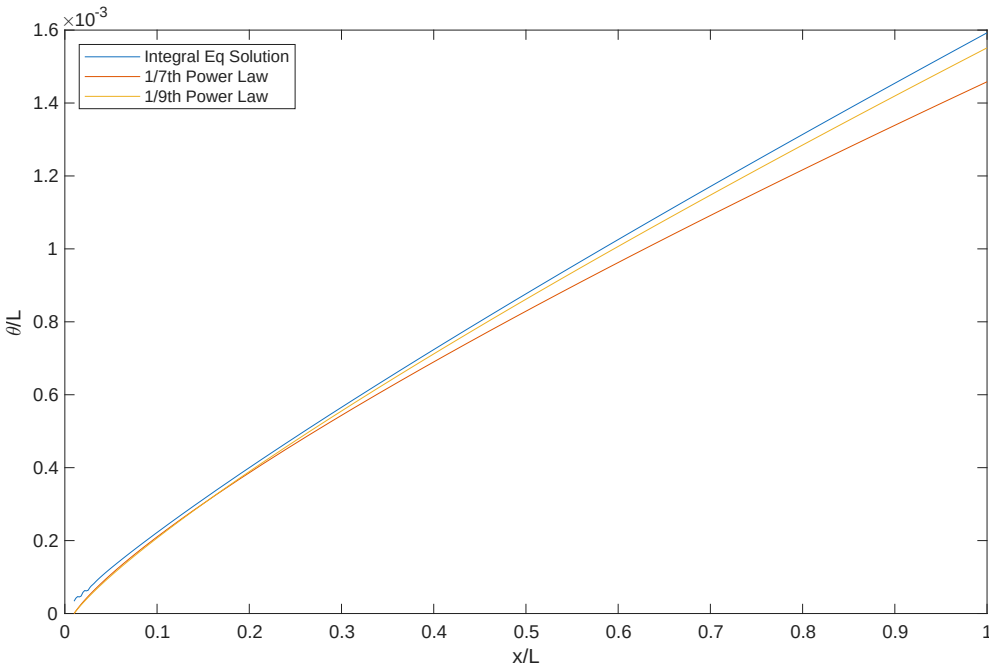


Figure 2: Turbulent boundary layer based on the integral equation for a flat plate with zero pressure gradient. The 1/7th and 1/9th power law are also plotted for comparison. (Exercise 4)

Listing 5: The function thickdash which evaluates $d\mathbf{y}/dx$ for each time interval. (Exercise 4)

```
function dthickdx = thickdash(xmx0, thick)
    global re_l dudx u0
    dthickdx = zeros(2,1);
    He = thick(2)/thick(1);
    if He>=1.46
        H =(11*He + 15)/(48*He-59);
    else
        H = 2.803;
    end
    ue = u0 + dudx*xmx0;
    re_thet = re_l * ue * thick(1);
    cf = 0.091416*((H-1)*re_thet)^(-0.232) * exp(-1.26*H);
    cdiss = 0.010026*((H-1)*re_thet)^(-1/6);
    dthickdx(1,1) = cf/2 - ((H+2)/ue)*dudx*thick(1);
    dthickdx(2,1) = cdiss - (3/ue)*dudx*thick(2);
end
```

Listing 6: The script for plotting a zero-pressure-gradient turbulent boundary layer using both integral method and the 1/7th and 1/9th power law. (Exercise 4)

```
clear;
close all;

global re_l dudx u0

re_l = 1e7;
dudx =  -0;
u0 = 1;

x0 = 0.01;
thick0(1) = 0.023*x0*(re_l*x0)^(-1/6);
thick0(2) = 1.83*thick0(1);
```

```
13
14 [delx, thickhist] =ode45(@thickdash, [0, 0.99], thick0);
15
16 theta7 = 0.037*delx.*(re_l.*(delx+x0)).^(-1/5);
17 theta9 = 0.023*delx.*(re_l.*(delx+x0)).^(-1/6);
18
19 figure;
20 plot(delx+x0, thickhist(:,1))
21 hold on;
22 plot(delx+x0, theta7)
23 hold off;
24 hold on;
25 plot(delx+x0, theta9)
26 hold off;
27 legend({'Integral Eq Solution', '1/7th Power Law', '1/9th Power Law'},'Location','northwest');
28 xlabel('x/L')
29 ylabel('\theta/L')
30 print -deps2c Week_2/Exercise4.eps;
```

5 Constant-velocity-gradient, turbulent boundary layer

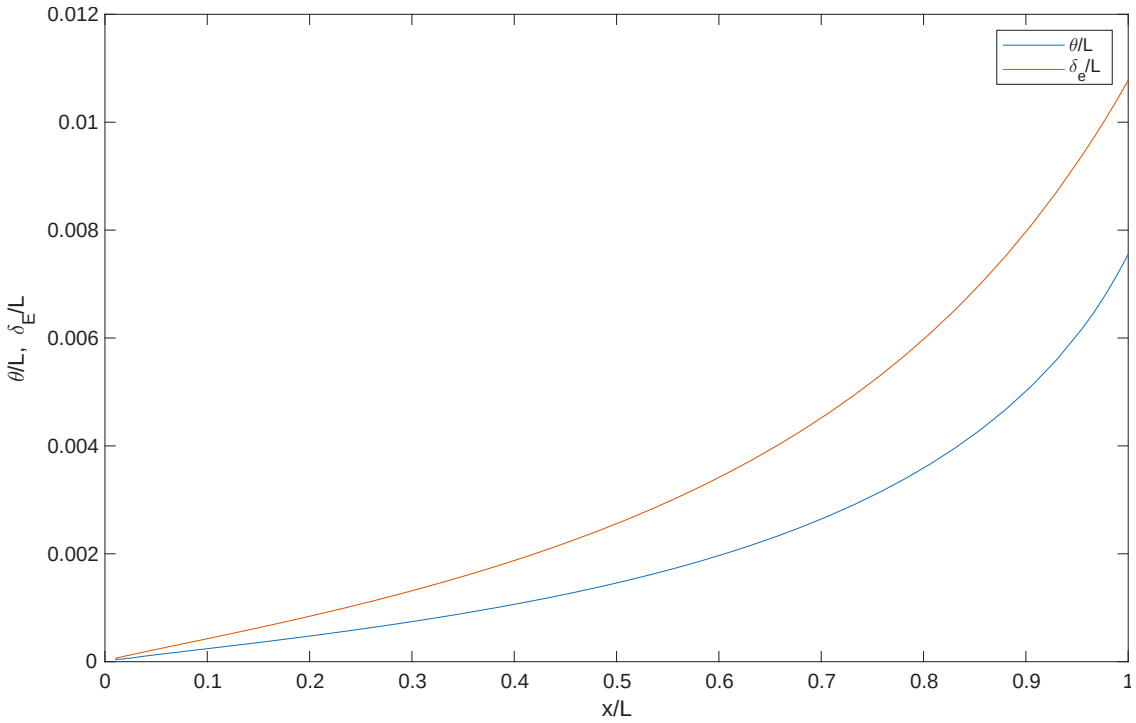


Figure 3: Momentum thickness and displacement thickness of a turbulent boundary layer with $Re_L = 1 \times 10^7$ and $du/dx = -0.5$. (Exercise 5)

du/dx	Separation Location, x/L
-0.25	No separation
-0.5	0.9887
-0.95	0.5230

Table 4: Separation locations for a turbulent boundary layer with $Re_L = 1 \times 10^7$ and a constant velocity gradient. (Exercise 5)

Re_L	Separation Location, x/L
1×10^6	0.8906
1×10^7	0.9887
1×10^8	No separation

Table 5: Separation locations for a turbulent boundary layer with $du/dx = -0.5$ and different Reynolds numbers.(Exercise 5)

Listing 7: Script for finding the location of turbulent separation points. (Exercise 5)

```
clear;
close all;


global re_l dudx u0

re_l = 1e6;
dudx =  -0.5;
u0 = 1;

x0 = 0.01;
```

```
thick0(1) = 0.023*x0*(re_l*x0)^(-1/6);
thick0(2) = 1.83*thick0(1);

[delx, thickhist] =ode45(@thickdash, [0, 0.99], thick0);

x=x0+delx;

all_He = thickhist(:,2) ./ thickhist(:,1);

sep_location=[];

for i = 1:length(all_He)
    if all_He(i) < 1.46 && isempty(sep_location)
        sep_location = i;
    end
end


disp(['Re_l ', num2str(re_l)]);
disp(['dudx ', num2str(dudx)]);
if ~isempty(sep_location)
    disp(['Separation at x/L = ', num2str(x(sep_location), '%.4f')]);
else
    disp('No separation before x/L = 1');
end



figure;
plot(x, thickhist(:,1), 'DisplayName', '\theta/L')
hold on;
plot(x,thickhist(:,2), 'DisplayName', '\delta_e/L' )
xlabel('x/L'); ylabel('\theta/L,  \delta_E/L');
legend;
%print -deps2c Week_2/Exercise5.eps;


figure;
plot(x, all_He);
xlabel('x/L'); ylabel('He');
yline(1.46, '--','Separation threshold');
```

6 Combined laminar and turbulent boundary layer evolution

To achieve turbulent separation at $x/L = 1$ with $Re_L = 1 \times 10^5$, it requires a velocity gradient of $du/dx = -0.38$.

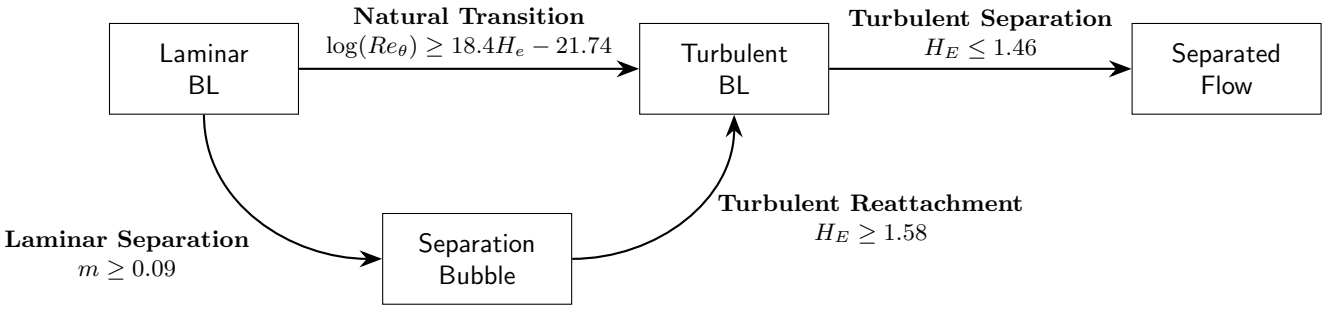
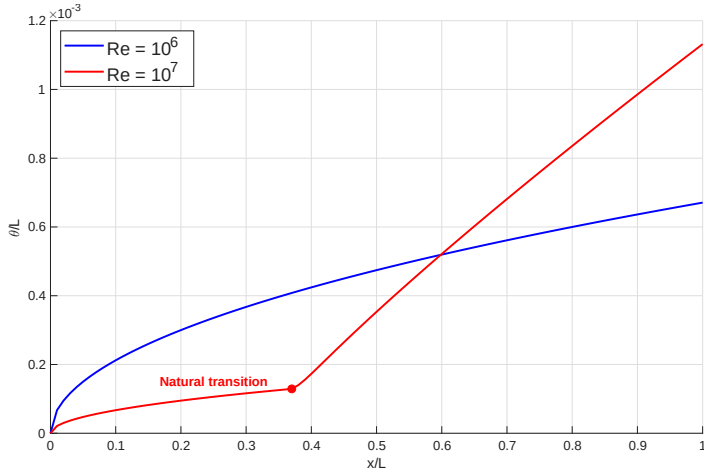
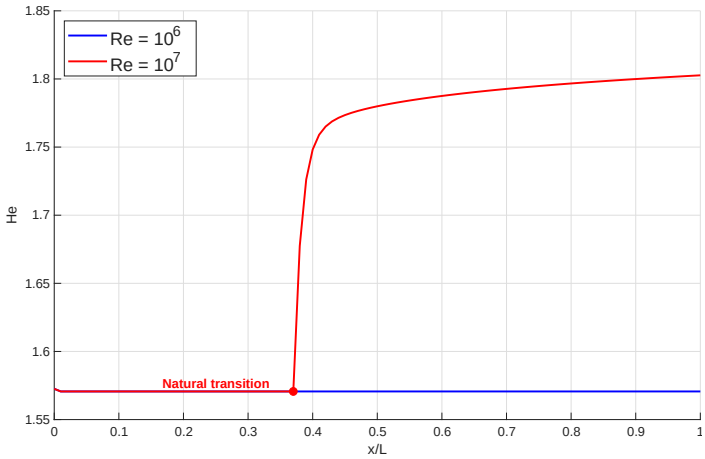


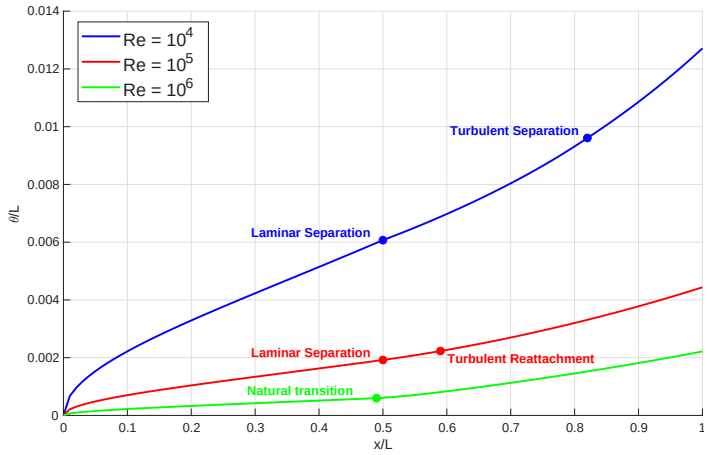
Figure 4: Flowchart describing the evolution of the boundary layer and the conditions for transitioning between regimes.



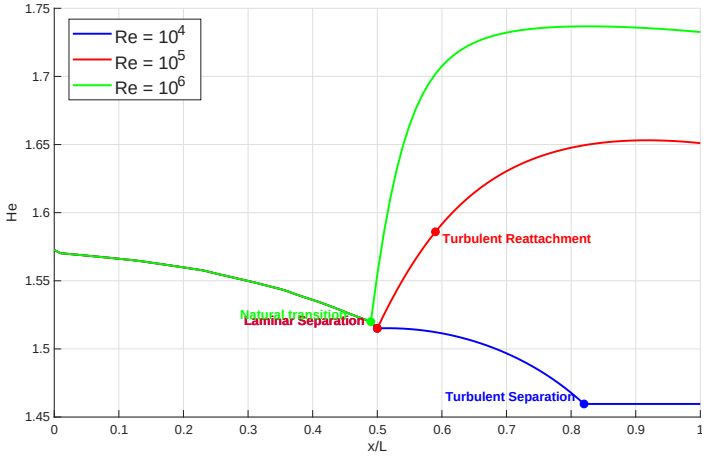
(a) Momentum thickness for zero-pressure-gradient.



(b) Energy shape factor, H_E, for zero-pressure-gradient.



(c) Momentum thickness for $du/dx = -0.25$.



(d) Energy shape factor, H_E, for $du/dx = -0.25$.

Figure 5: Plots of momentum thickness and energy shape factor against distance for zero-pressure-gradient, and two values of Reynolds number. (Exercise 6)

Listing 8: Function to calculate the boundary layer for a fixed Reynolds number and velocity gradient. (Exercise 6)

```
function [theta, Hes, int, ils, itr, its,x] = calc_bl(dudx, re_l)
    global re_l dudx u0

    np = 101;
    %for turbulent separation at x/L=1, dudx = -0.383
    x = linspace(0,1,np); %length varies from 0->1

    theta = zeros(1,np);  %theta squared at each position
    Hes = zeros(1,np);
    Hes(1) = 1.57258;


    laminar = true;
    integ = 0;
    int = 0; %location of laminar->turbulent transition
    ils = 0; %location of laminar separation
    itr = 0; %location of turbulent reattachment
    its = 0; %location of turbulent separation
    i=1;
    u_ = linspace(1, 1+dudx, np);
    while laminar && i<np
        i=i+1;
        integ = integ + ueintbit(x(i-1), u_(i-1), x(i), u_(i));
        theta(i) = sqrt((0.45/re_l) * u_(i)^(-6) * integ);
        m = -re_l * theta(i)^2  * dudx;
        H = thwaites_lookup(m);
        Hes(i) = laminar_He(H);
        re_thet = re_l * u_(i) * theta(i);
        if log(re_thet) >= 18.4*Hes(i) - 21.74
            laminar = false;    %transition
            int = i;

        elseif m>= 0.09
            laminar = false;
            ils = i;
            Hes(i) = 1.51509;
        end
    end
    delta_e = Hes(i)*theta(i);

    while its == 0 && i<np %loop for turbulent boundary layer
        i=i+1;
        thick0 = [theta(i-1), delta_e];
        u0 = u_(i-1); %velocity at start of panle
        [delx, thickhist] = ode45(@thickdash, [0, x(i)-x(i-1)], thick0);
        theta(i) = thickhist(end, 1);
        delta_e = thickhist(end, 2);
        Hes(i) = delta_e/theta(i);
        if Hes(i)>1.58 && ils ~= 0 &&itr==0
            itr = i; %turbulent reattachment
        end
        if Hes(i) <1.46
            its = i; %turbulent separation
        end
    end

    if Hes(i)>=1.46
        H =(11*Hes(i) + 15)/(48*Hes(i)-59);
    else
        H = 2.803;
    end
    if its ~= 0
        Hes(its+1:np) = Hes(i);
```

```
        theta(its+1:np) = theta(i).*(u_(i)./u_(i+1:np)).^(H+2);
    end

end
```

Listing 9: Script for plotting momentum thickness and energy shape factor with annotations for regime transitions.

```
clear;
close all;
global re_l dudx
dudx = 0 ;

re_l = 1e6;
[theta1, hes1, int1, ils1, itr1, its1,x] = calc_bl(dudx, re_l);

re_l = 1e7;
[theta2, hes2, int2, ils2, itr2, its2, x] = calc_bl(dudx, re_l);

figure(1);
hold on;
p1 = plot(x, theta1, 'b-', 'LineWidth', 1.5, 'DisplayName', 'Re = 10^6');
p2 = plot(x, theta2, 'r-', 'LineWidth', 1.5, 'DisplayName', 'Re = 10^7');
add_bl_labels(x, theta1, int1, ils1, itr1, its1, 'b');
add_bl_labels(x, theta2, int2, ils2, itr2, its2, 'r');
hold off;

xlabel('x/L');
ylabel('\theta/L');
lgd = legend([p1, p2], 'Location', 'best');
lgd.FontSize = 14;
lgd.Location = northwest;
grid on;
print -deps2c Week_2/Exercise6a.eps;

figure(2);
hold on;
p1 = plot(x, hes1, 'b-',  'LineWidth', 1.5, 'DisplayName', 'Re = 10^6');
p2 = plot(x, hes2, 'r-',  'LineWidth', 1.5, 'DisplayName', 'Re = 10^7');
add_bl_labels(x, hes1, int1, ils1, itr1, its1, 'b');
add_bl_labels(x, hes2, int2, ils2, itr2, its2, 'r');
hold off;

xlabel('x/L');
ylabel('He');
lgd = legend([p1, p2], 'Location', 'best');
lgd.FontSize = 14;
lgd.Location = northwest;
grid on;
print -deps2c Week_2/Exercise6b.eps;


%now add velocity gradient
dudx = -0.25;

re_l=1e4;
[theta1, hes1, int1, ils1, itr1, its1,x] = calc_bl(dudx, re_l);

re_l=1e5;
[theta2, hes2, int2, ils2, itr2, its2, x] = calc_bl(dudx, re_l);

re_l=1e6;
[theta3, hes3, int3, ils3, itr3, its3, x] = calc_bl(dudx, re_l);

figure(3);
```

```
hold on;
p1 = plot(x, theta1, 'b-', 'LineWidth', 1.5, 'DisplayName', 'Re = 10^4');
p2 = plot(x, theta2, 'r-', 'LineWidth', 1.5, 'DisplayName', 'Re = 10^5');
p3 = plot(x, theta3, 'g-', 'LineWidth', 1.5, 'DisplayName', 'Re = 10^6');
add_bl_labels(x, theta1, int1, ils1, itr1, its1, 'b');
add_bl_labels(x, theta2, int2, ils2, itr2, its2, 'r');
add_bl_labels(x, theta3, int3, ils3, itr3, its3, 'g');
hold off;

xlabel('x/L');
ylabel('\theta/L');
lgd = legend([p1, p2,p3], 'Location', 'best');
lgd.FontSize = 14;
grid on;
print -deps2c Week_2/Exercise6c.eps;

figure(4);
hold on;
p1 = plot(x, hes1, 'b-',  'LineWidth', 1.5, 'DisplayName', 'Re = 10^4');
p2 = plot(x, hes2, 'r-',  'LineWidth', 1.5, 'DisplayName', 'Re = 10^5');
p3 = plot(x, hes3, 'g-',  'LineWidth', 1.5, 'DisplayName', 'Re = 10^6');
add_bl_labels(x, hes1, int1, ils1, itr1, its1, 'b');
add_bl_labels(x, hes2, int2, ils2, itr2, its2, 'r');
add_bl_labels(x, hes3, int3, ils3, itr3, its3, 'g');
hold off;

xlabel('x/L');
ylabel('He');
lgd = legend([p1, p2,p3], 'Location', 'best');
lgd.FontSize = 14;
grid on;
print -deps2c Week_2/Exercise6d.eps;



function add_bl_labels(x, y, i_nt, i_ls, i_tr, i_ts, color)
    if i_nt ~= 0
        %natural transition annotation
        plot(x(i_nt), y(i_nt), [color 'o'], 'MarkerFaceColor', color);
        text(x(i_nt), y(i_nt), '    Natural transition          ', 'Color', color, 'FontWeight', 'bold', '
    VerticalAlignment', 'bottom', 'HorizontalAlignment','right');
    end

    if i_ls ~= 0
        %laminar separation annotation
        plot(x(i_ls), y(i_ls), [color 'o'], 'MarkerFaceColor', color, 'HandleVisibility', 'off');
        text(x(i_ls), y(i_ls), '  Laminar Separation      ', 'Color', color, 'FontWeight', 'bold', '
    VerticalAlignment', 'bottom', 'HorizontalAlignment','right');
    end

    if i_tr ~= 0
        %turbulent reattachment annotation
        plot(x(i_tr), y(i_tr), [color 'o'], 'MarkerFaceColor', color);
        text(x(i_tr), y(i_tr), '  Turbulent Reattachment       ', 'Color', color, 'FontWeight', 'bold', '
    VerticalAlignment', 'top', 'HorizontalAlignment','left');
    end

    if i_ts ~= 0
        %turbulent separation annotation
        plot(x(i_ts), y(i_ts), [color 'o'], 'MarkerFaceColor',color);
        text(x(i_ts), y(i_ts), '  Turbulent Separation     ', 'Color', color, 'FontWeight', 'bold', '
    VerticalAlignment', 'bottom', 'HorizontalAlignment','right');
    end
end
```